Projective measurement for a qubit

$d=2$

$|0\rangle\langle 0| = \frac{I+Z}{2}$ $\quad |+\rangle\langle +| = \frac{I+X}{2}$

$|1\rangle\langle 1| = \frac{I-Z}{2}$ $\quad |-\rangle\langle -| = \frac{I-X}{2}$

$|+y\rangle\langle +y| = \frac{I+Y}{2}$

$|-y\rangle\langle -y| = \frac{I-Y}{2}$

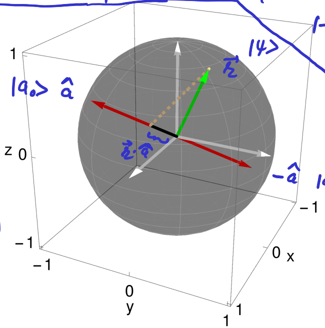
State $\rho = |\psi\rangle\langle\psi| = \dfrac{I + \vec{r}\cdot\vec{\sigma}}{2}$ $\qquad \Pi^2 = \Pi$

$\Pi_0 + \Pi_1 = I$

$\mathcal{B}_a = \{|a_0\rangle, |a_1\rangle\}$

Projector $\Pi_j = |a_j\rangle\langle a_j| = \dfrac{I + (-1)^j\,\widehat{a}\cdot\vec{\sigma}}{2} = \dfrac{I \pm \hat{a}\cdot\vec{\sigma}}{2}$

$(\hat{a}\cdot\vec{r})I + i\,\underline{(\hat{a}\times\vec{r})\cdot\vec{\sigma}}$



Outcome $j \in \{0, 1\}$

$\Pi_0\rho = \left(\frac{I+\hat{a}\cdot\vec{\sigma}}{2}\right)\left(\frac{I+\vec{r}\cdot\vec{\sigma}}{2}\right) = \frac{1}{4}\left[I + \hat{a}\cdot\vec{\sigma} + \vec{r}\cdot\vec{\sigma} + \overbrace{(\hat{a}\cdot\vec{\sigma})(\vec{r}\cdot\vec{\sigma})}\right]$

Born Rule

Probability $p_j = \mathrm{tr}(\Pi_j\,\rho) = |\langle a_j|\psi\rangle|^2 = \dfrac{1 + (-1)^j\,\widehat{a}\cdot\vec{r}}{2} = \dfrac{1 \pm \hat{a}\cdot\vec{r}}{2} \geq 0$

$p_0 + p_1 = 1$

Expectation value $\langle\widehat{a}\cdot\vec{\sigma}\rangle = \widehat{a}\cdot\vec{r} = (+1)\underbrace{\langle\Pi_0\rangle}_{p_0} + (-1)\underbrace{\langle\Pi_1\rangle}_{p_1} = p_0 - p_1 = \hat{a}\cdot\vec{r}$

$\hat{a}\cdot\vec{\sigma}\,|a_j\rangle = (-1)^j\,|a_j\rangle$

video

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Symmetric Informationally Complete POVM for a qubit

$d^2 = 4$

quantum state estimation

State $\rho = \dfrac{I + \vec{r}\cdot\vec{\sigma}}{2}$ $\quad \vec{r} = (x, y, z)$

$\Pi^2 \neq \Pi$

$p_0 + p_1 = 1$

Weighted Projector $\Pi_j = \frac{1}{2}\underbrace{|a_j\rangle\langle a_j|} = \dfrac{I + \hat{a}_j\cdot\vec{\sigma}}{4}$

$\hat{a}_0 = \frac{1}{\sqrt{3}}(1, 1, 1)$

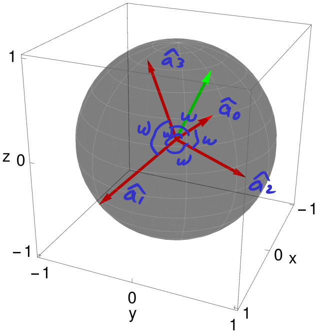
$\hat{a}_1 = \frac{1}{\sqrt{3}}(1, -1, -1)$

$\hat{a}_2 = \frac{1}{\sqrt{3}}(-1, 1, -1)$

$\hat{a}_3 = \frac{1}{\sqrt{3}}(-1, -1, 1)$

SIC-POVM $= \{\Pi_0 \cdots \Pi_3\}$

$\hat{a}_i \cdot \hat{a}_j = -\frac{1}{3}$ for $i \neq j$



Outcome $j \in \{0, 1, 2, 3\}$

Born Probability $p_j = \mathrm{tr}(\Pi_j\,\rho) = \dfrac{1 + \hat{a}_j\cdot\vec{r}}{4}$ ≥ 0

Constraints:

$$\sum_j \hat{a}_j = \vec{0} \leftrightarrow \sum_j \Pi_j = I \longrightarrow \sum_j p_j = 1 \text{ due to } \mathrm{tr}(\rho) = 1$$

$$\sum_j p_j^2 \leq \tfrac{1}{3} \text{ due to } \underline{\mathrm{tr}(\rho^2) \leq 1}$$

$0 \leq \rho$

video

Universal set of "discrete" single-qubit gates

$R_{\hat{\eta}}(t) = R_z(\alpha)\, R_y(\beta)\, R_z(\xi)$

$\hat{\eta}, t$ are fix

z, y

$\alpha \beta \xi \in \mathbb{R}$

- A fault-tolerant quantum computation works only for a **discrete** set of gates.
- A discrete set of (finite no. of) gates such as $\{H, T\}$ can not be (can be) used to implement an arbitrary unitary operation U exactly (approximately), since the set of unitary operations is continuous.

$H \to \hat{\eta} = \frac{1}{\sqrt{2}}(1,0,1)$ $\quad t = \pi$

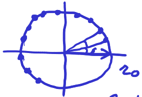
$T \to \hat{\eta} = (0,0,1)$ $\quad t = \pi/4$

- **The Solovay–Kitaev theorem:** For a given accuracy $\epsilon > 0$, it is possible to approximate any single-qubit unitary gate U using $m \approx O(\log^c(\frac{1}{\epsilon}))$ gates from a certain finite set of gates, where $c \approx 2$.

$HZH = X$

$$R_\eta(\theta) = T\,HTH = \Big[\cos(\tfrac{\pi}{8})\,I - \mathrm{i}\sin(\tfrac{\pi}{8})\,Z\Big]\Big[\cos(\tfrac{\pi}{8})\,I - \mathrm{i}\sin(\tfrac{\pi}{8})\,X\Big] = \underbrace{\left(\cos\frac{\pi}{8}\right)^2}_{\cos\frac{\theta}{2}} I - \mathrm{i}\sin(\tfrac{\pi}{8})\,\vec{\eta}\cdot\vec{\sigma}$$

$\vec{\eta} = \left[\cos\frac{\pi}{8},\ \sin\frac{\pi}{8},\ \cos\frac{\pi}{8}\right]$



$R(\delta)^m$

$R(m\delta')$

$$\theta = 2\arccos\left(\frac{\sqrt{2}+1}{2\sqrt{2}}\right) \text{ is an irrational multiple of } 2\pi.$$

$|\psi_{out}\rangle$ $\quad |\psi_{out}^{app}\rangle$

$ZX = iY$

$\delta = \frac{p}{q}2\pi$

$q\delta = p\,2\pi$

$\delta \neq \frac{p}{q}2\pi$

$$\text{Error} := \max_{|\psi\rangle}\ \text{Dist}\big(R_\eta(t)|\psi\rangle\,,\ \underbrace{(R_\eta(\theta))^m}_{R_\eta(m\theta)}|\psi\rangle\big) < \tfrac{\epsilon}{3}$$

$$R_{\eta'}(\theta) = HR_\eta(\theta)H = HTHT = H\,[THT\underbrace{H]\,H}_{I}$$

$HXH = Z$

$HYH = -Y$

$HZH = X$

$\vec{\eta}' = \left[\cos\pi/8,\ -\sin\frac{\pi}{8},\ \cos\frac{\pi}{8}\right]$

$$U \approx \overbrace{\underbrace{R_\eta(\theta)^{m_1}}_{R_\eta(m_1\theta)}\ \underbrace{R_{\eta'}(\theta)^{m_2}}_{R_{\eta'}(m_2\theta)}\ \underbrace{R_\eta(\theta)^{m_3}}_{R_\eta(m_3\theta)}}^{U_{app}}$$

$\text{Error}(U, U_{app}) < \epsilon$

M. A. Nielsen and I. L. Chuang, *Quantum Compu* *and Quantum Information*, Chapt 4 and Appendix 3.